



**JABATAN PEMBANGUNAN KEMAHIRAN
KEMENTERIAN SUMBER MANUSIA
ARAS 7 & 8 BLOK D4, KOMPLEKS D
PUSAT PENTADBIRAN KERAJAAN PERSEKUTUAN
62530 PUTRAJAYA**

KERTAS PENERANGAN (*INFORMATION SHEET*)

KOD DAN NAMA PROGRAM / <i>PROGRAM CODE AND NAME</i>	Z-009-3:2015 CORE ABILITIES	
TAHAP / LEVEL	3 (THREE)	
NO. DAN TAJUK KEBOLEHAN TERAS / <i>CORE ABILITY NO. AND TITLE</i>	CA02 INFORMATION TECHNOLOGY AWARENESS	
NO. DAN PENYATAAN KEBOLEHAN / <i>ABILITY NO. AND STATEMENT</i>	02.01 AWARE OF INFORMATION TECHNOLOGY UPDATE 02.02 AWARE OF INFORMATION TECHNOLOGY REGULATORY	
NO. KOD / CODE NO.	Z-009-3:2015-M02/P(1/2)	Mukasurat / Page : 1 Drpd / of : 10

TAJUK/TITLE : IT UPDATE AWARENESS

TUJUAN/PURPOSE :

This Information Sheet is intended to describe on the Information Technology Update Awareness of IT operation within the organization. The person who is competent in this CA shall be able to infuse and aware of information technology update in daily work.

PENERANGAN/INFORMATION :

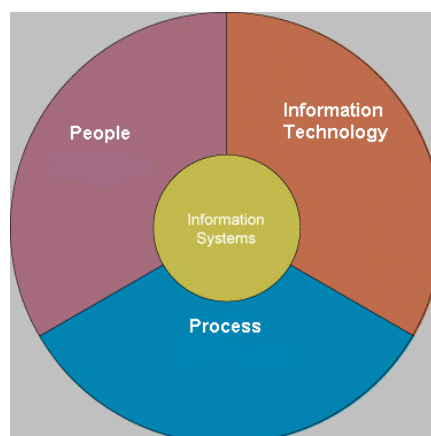
1. Definition

1.1 **Information technology (IT)** is the application of computers and telecommunications equipment to store, retrieve, transmit and manipulate data, often in the context of a business or other enterprise.



Picture 1.1 . Information Technology Relationship

1.2 An **Information System (IS)** is a system composed of people and computers that processes or interprets information. The term is also sometimes used in more restricted senses to refer to only the software used to run a computerized database or to refer to only a computer system.



Piechart 1.2 Information Systems Relationship Chart

1.3 **Information systems** is also basically an academic study of systems with a specific reference to information and the complementary networks of hardware and software that people and organizations use to collect, filter, process, create and also

distribute data.

1.4 Any specific information system aims to support operations, management and decision-making. An information system is the information and communication technology (ICT) that an organization uses, and also the way in which people interact with this technology in support of business processes.

1.5 **Technology literacy** is the ability of an individual, working independently and with others, to responsibly, appropriately and effectively use technology tools to access, manage, integrate, evaluate, design, create and communicate information. It is a lifelong knowledge and skills in the 21st century.

2. Components of Information System Technologies

2.1 Generally there are 6 components of Information System Technologies. They are :

i) Hardware

The term hardware refers to machinery. This category includes the computer itself, which is often referred to as the central processing unit (CPU), and all of its support equipments. Among the support equipments are input and output devices, storage devices and communications devices.

ii) Software

The term software refers to computer programs and the manuals (if any) that support them. Computer programs are machine-readable instructions that direct the circuitry within the hardware parts of the CBIS to function in ways that produce useful information from data. Programs are generally stored on some input / output medium, often a disk or tape.

iii) Data

Data are facts that are used by programs to produce useful information. Like programs, data are generally stored in machine-readable form on disk or tape until the computer needs them.

iv) Procedures

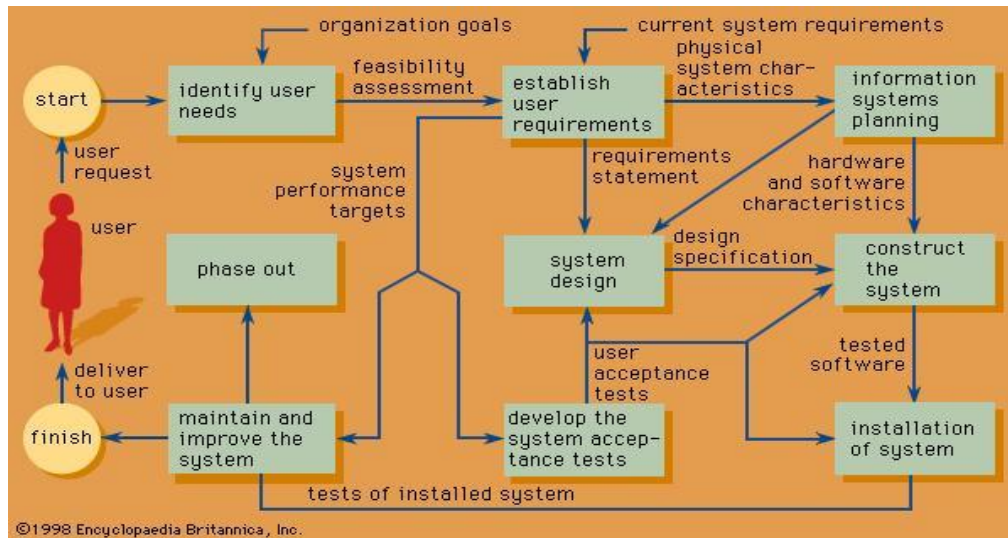
Procedures are the policies that govern the operation of a computer system. "Procedures are to people what software is to hardware" is a common analogy that is used to illustrate the role of procedures in a CBIS.

v) Networks

Are a connecting system that allows diverse computers to distribute resources.

vi) People

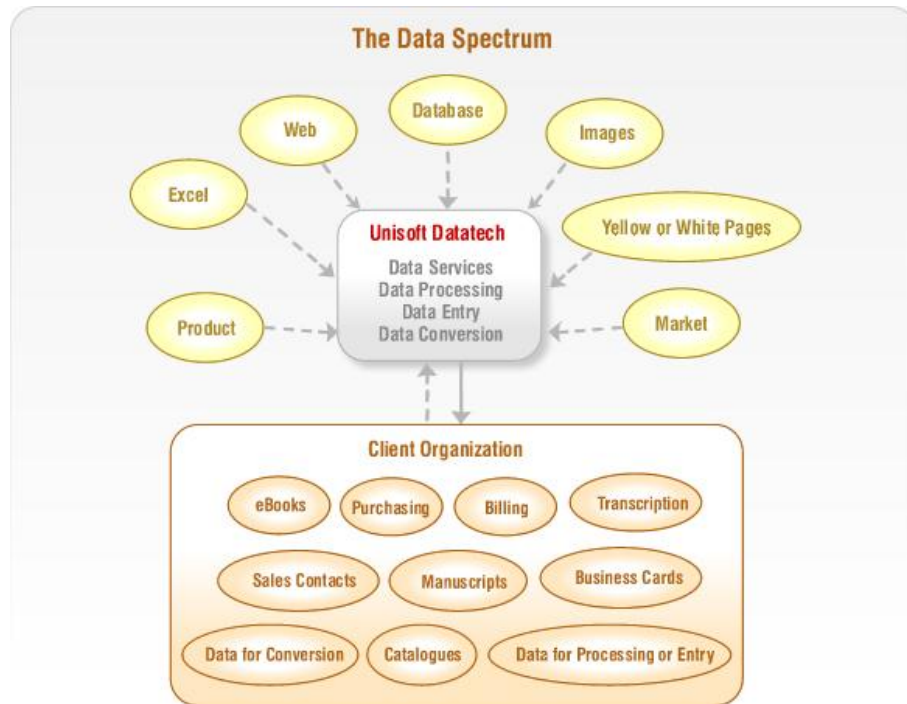
Every Computer Based Information System (CBIS) needs people if it is to be useful. Often the most over-looked element of the CBIS is the people, probably the component that most influence the success or failure of information systems.



Flowchart 2.1 Sample of Networking Flow

3. Data entry

3.1 Data entry refers to **user actions involving input of data to a computer, and computer responses to such inputs**. The simplest kind of data entry consists merely of pointing at something -- selecting an item or designating a position on a computer-generated display. In more complicated modes of data entry, a user may have to control the format of data inputs as well as their contents. Thus questions of format control in text entry/editing and graphic interaction may properly be considered questions of data entry.

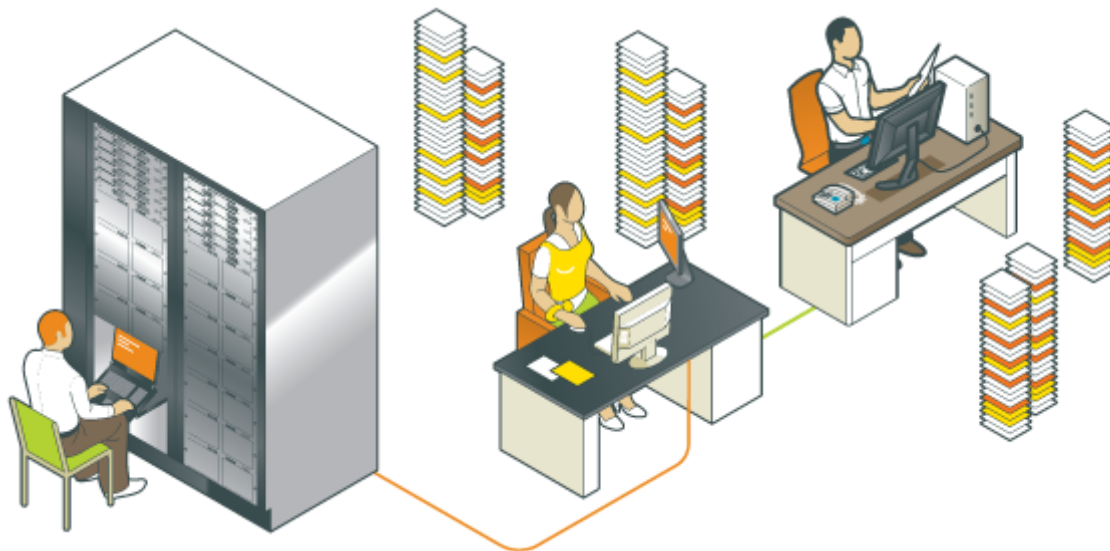


Flowchart 3.1 Sample Data Spectrum by Unisoft Datatech

3.2 In the world of data entry, there are several software available such as Microsoft Access, Snappii, Vikingsoft, Peoplesoft, Quickbooks, Microsoft Excel, Visio, SAP and others. The requirements of software selection vary from individuals to companies as each software give different abilities that suits their need. In all commerce/business entity, data entry is crucial as it keeps in a systematic matter the transaction history, customer database, purchase records, official instruction/request, financial flow and track record, budget flow, work instruction etc...etc.. All these must be securely but manageably easy to be worked with and easy to be retrieved by authorized personnel



Picture 3.2.a Sample of Softwares



Picture 3.2.b. Data Entry Basic Sample Flow

4. What are the basic considerations in selecting the right software?

4.1 There are **9** considerations in selecting the right software. They are:

- i) Features against requirements – most software offers more than what is required thus future-proofing is a good strategy if the cost is right
- ii) Cost/benefit analysis :
 - a) Initial setup cost
 - b) (Main) ongoing charge–monthly or annual (might depend on specific modules)
 - c) Per-member surcharge
 - d) Per-transaction charges (% or \$)
 - e) Technical support fees
 - f) Update charges
- iii) Future upgradability – All software normally have their update cycle and sometimes offer upgrading. One of the things need to be confirm is the planning of the software maker for the future; either they will only upgrade the software or totally replace it with a new programming. This will impact the cost versus functionality.
- iv) Hardware compatibility – current specification vs. compatible specification
- v) System compatibility – Does the software works in the Operating System currently used?If it works,how well does it operate in the Operating System?

NO. KOD / CODE NO.	Z-009-3:2015-M02/P(1/2)	Muka Surat / Page : 7 Drpd / of : 10
---------------------------	--------------------------------	---

- vi) Customer support and vendor reliability - This is critical as it defines the relationship between vendor/seller/software maker and buyer. This affects warranty, technical services and cost. When the software malfunctions for whatever reason, user needs to refer to the expert to find solutions to it
- vii) Security – The level of security of the software against intruder is highly important as to avoid valuable data. It also should cover hacking and virus protection.
- viii) Staff and learning curve (Ease of use) – Person responsible and authorized to use the software must be given proper training to use the software efficiently. The ease of use is also a factor to consider as it will impact the overall efficiency of the process.
- ix) Reporting – Each process should be able to produce a report/summary in a format deemed sufficient to the user.

5. Why do we need Technology Update Awareness?

5.1 Interaction between users and technology/systems is unavoidable. One needs to be aware of the connection and advantages/disadvantages directly or indirectly affecting his work. He/she needs to be aware of the current IT trending and any updates related to his work. Technology improves production/communication and database handling. Obsolete technology can affect efficiency and production. As one moves up in the career ladder, sophistication in work and interaction with its surrounding increases thus justifying the needs to be aware and updated with technology.

5.2 Previously, all paperworks, reports, charts, presentations, minutes of meetings etc...etc.. are done manually and fully utilizing the usage of paper. When the transition from paper to paperless occurs, i.e. using computer/software, one needs to know how to transform the format and information into a softcopy version. For example, if a report previously prepared using typewriter, when converting it into a softcopy version, one needs to know how to use the software and how to format the report. Using software, there are endless selections of options to present the report. These softwares are usually updated and continuously evolving based on continuous changes thus user needs to know the updates and how they affect them. This technology update awareness applies to all type/level of technology.

NO. KOD / CODE NO.	Z-009-3:2015-M02/P(1/2)	Muka Surat / Page : 8 Drpd / of : 10
---------------------------	--------------------------------	---

5.3 Examples of the areas needed to be aware of are as such:

- i) How to use the software and its capabilities
- ii) Knowledge of suitable IT component/systems that optimizes work and delivery
- iii) Which machine/computers/system/software that is the optimum solution
- iv) What/who are the current market trends/users
- v) Cost to solution

NO. KOD / CODE NO.	Z-009-3:2015-M02/P(1/2)	Muka Surat / Page : 9 Drpd / of : 10
---------------------------	--------------------------------	---

SOALAN/QUESTION:

1. What is the relationship between Information Technology and Information System?

.....

.....

2. What are the 6 components of Information System Technology

.....

.....

3. Why do we need Technology Update Awareness?

.....

.....

4. What is data entry?

.....

.....

5. How many considerations needed in selecting correct software?

.....

.....

NO. KOD / CODE NO.	Z-009-3:2015-M02/P(1/2)	Muka Surat / Page : 10 Drpd / of : 10
---------------------------	--------------------------------	--

RUJUKAN / REFERENCES:

1. <http://www.montgomeryschoolmd.org/departement/techlit/docs.htm>
2. http://www.wikipedia.org/wiki/information_technology
3. http://www.wikipedia.org/wiki/information_system
4. <http://www.coloradotechliteracy.org/org/documentation/module1/definition.htm>
5. http://www.uotechnology.edu.iq/ce/Lectures/SarmadFuad-MIS/MIS_Lecture_3.pdf
6. <http://www.umsl.edu/~joshik/msis480/chapt02.htm>
7. <http://global.britannica.com/topic/information-system>
8. http://www.nutrientdataconf.org/pastconf/ndbc19/7-1_westrich.pdf
9. <http://knowledge.wildapricot.com/display/technology/Defining+Software+Requirements>